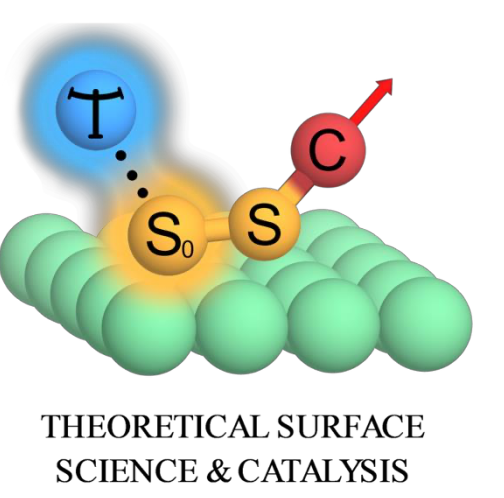
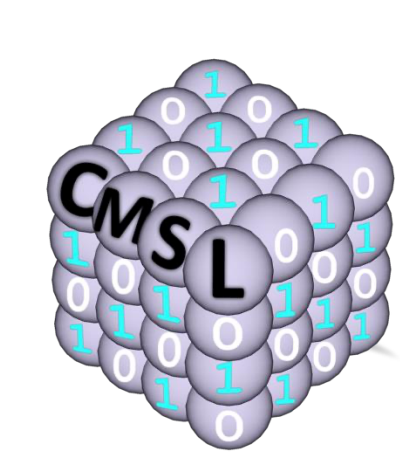
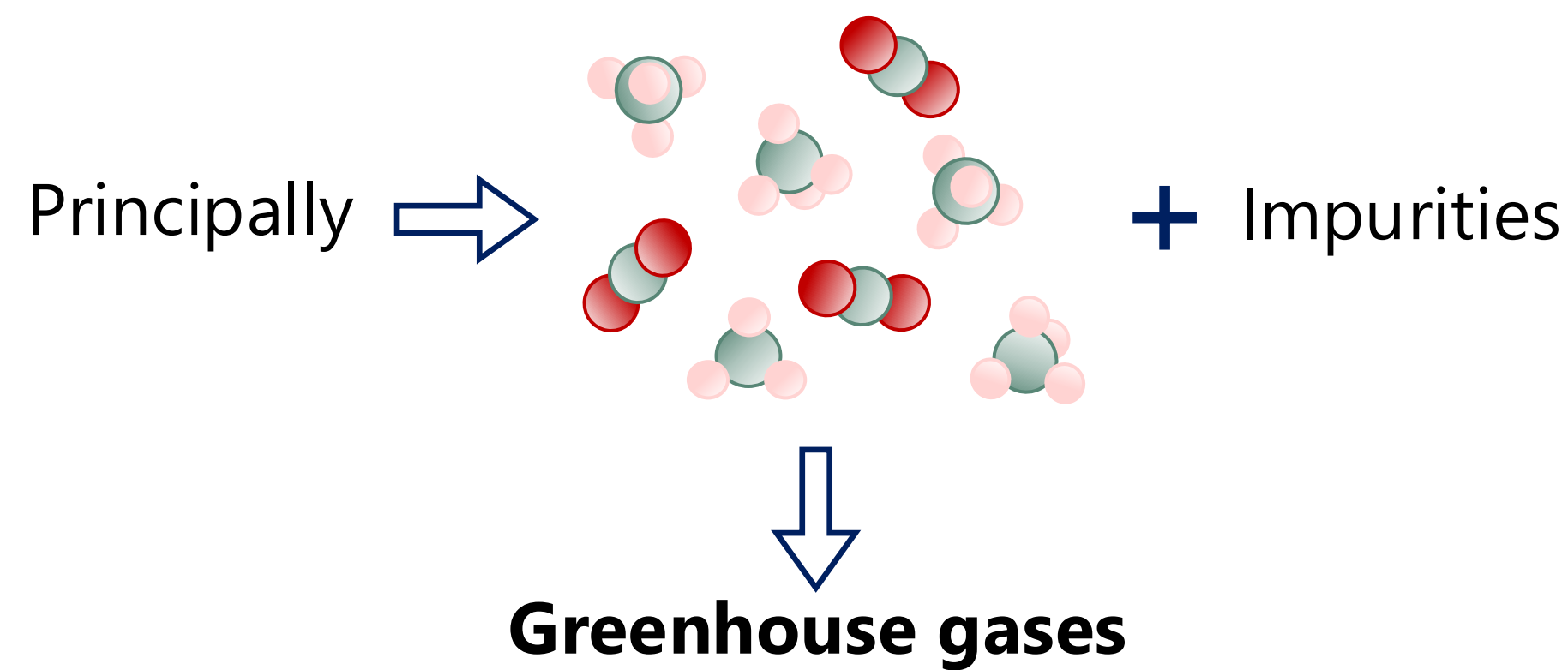




Introduction

Biogas: fuel derived from the anaerobic digestion of organic matter



5% more **CO₂** than CH₄ in the atmosphere

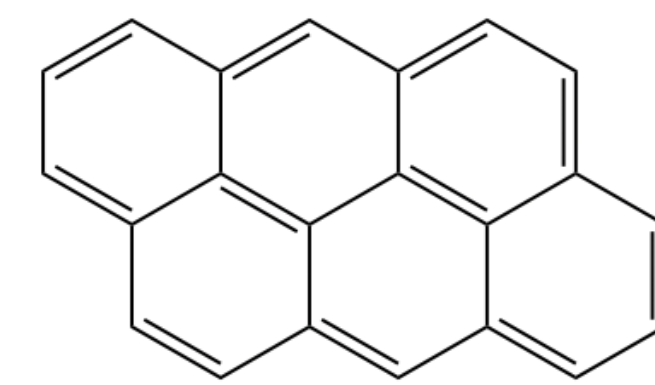
CH₄ retains heat **23 times** more than CO₂ does

Burning CH₄ would de facto **reduce its greenhouse gas impact**. Biogas streams feature **50- 75% (v/v) of CH₄** purity admixed with **25-50% of CO₂**. CH₄ requires a purity of **>95% (v/v)** to be properly used as fuel, which asks for **biogas upgrading** processes to separate CO₂ from CH₄ before its use as fuel.

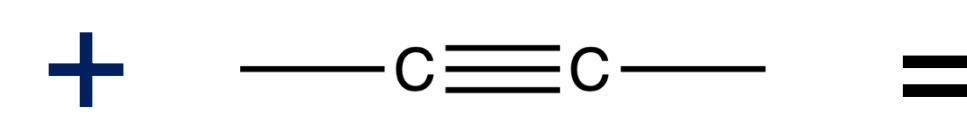
Results

What are grazynes?

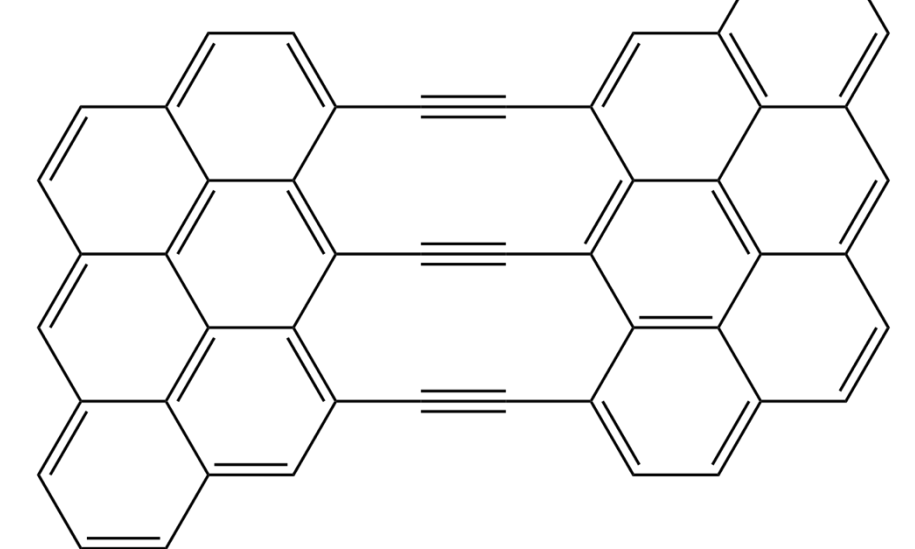
Graphene stripes



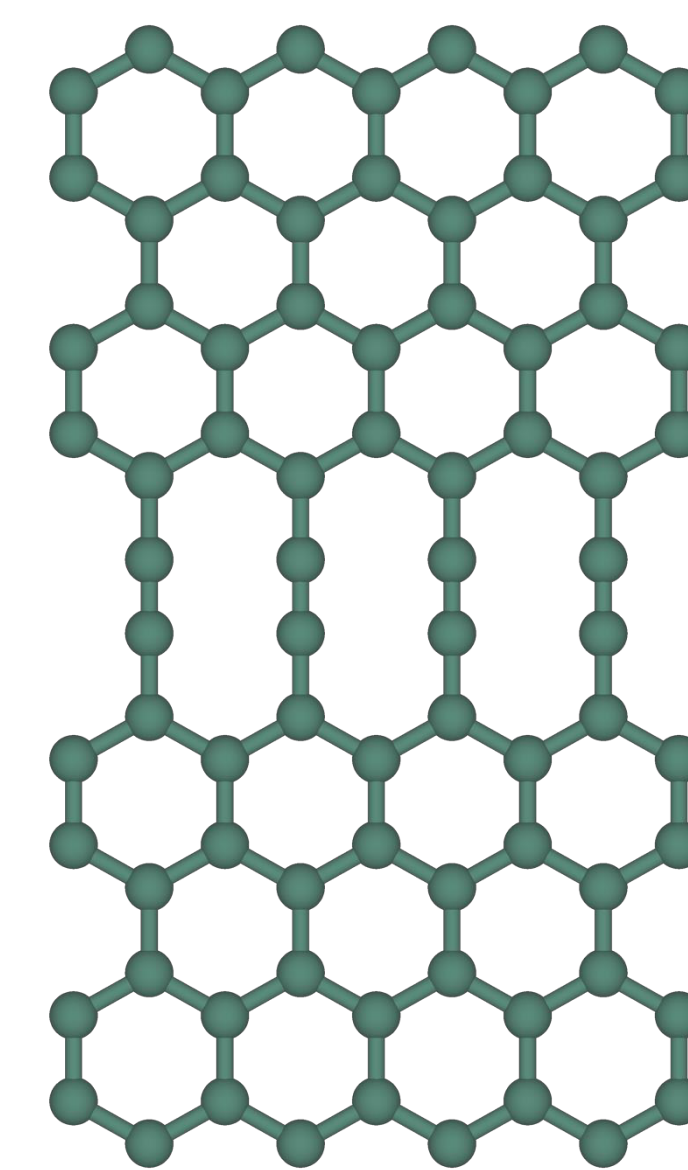
Acetylenic Linkages



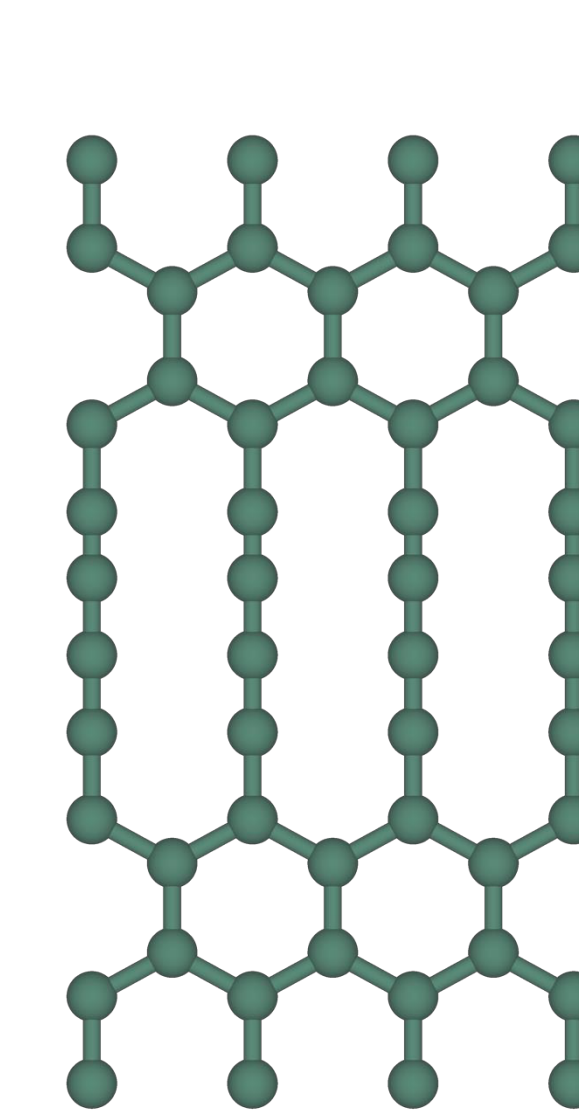
Grazyne¹



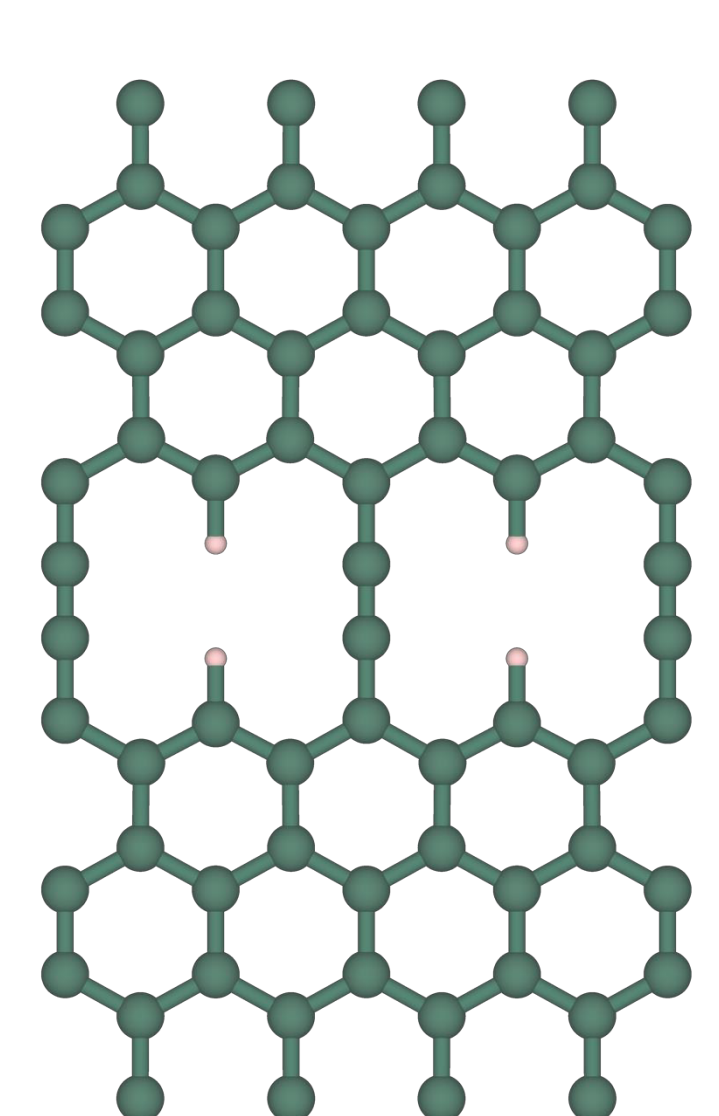
 **They can be modified in different ways** 



Modifying the graphene **width**

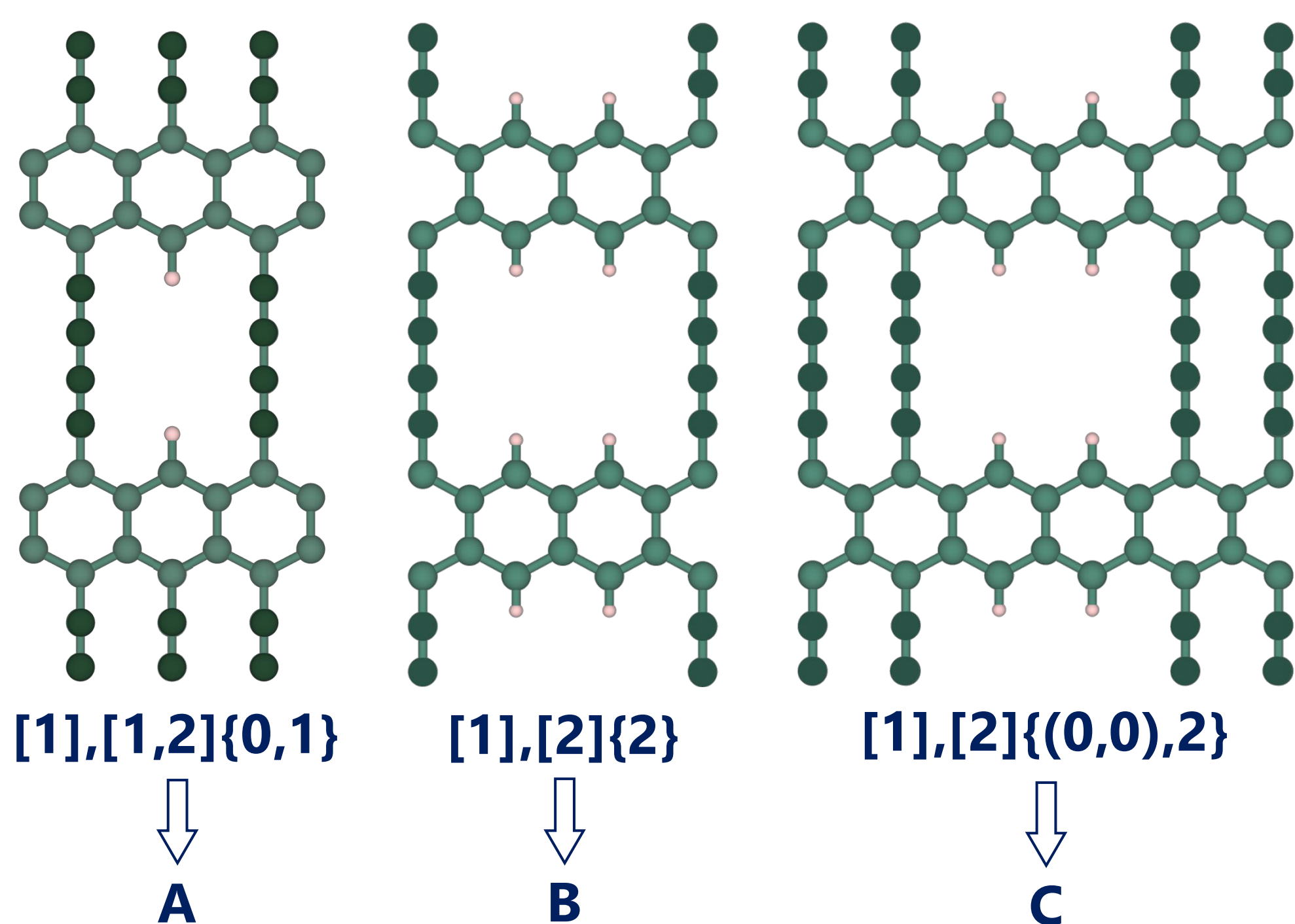


Enlarging acetylenic **linkages**

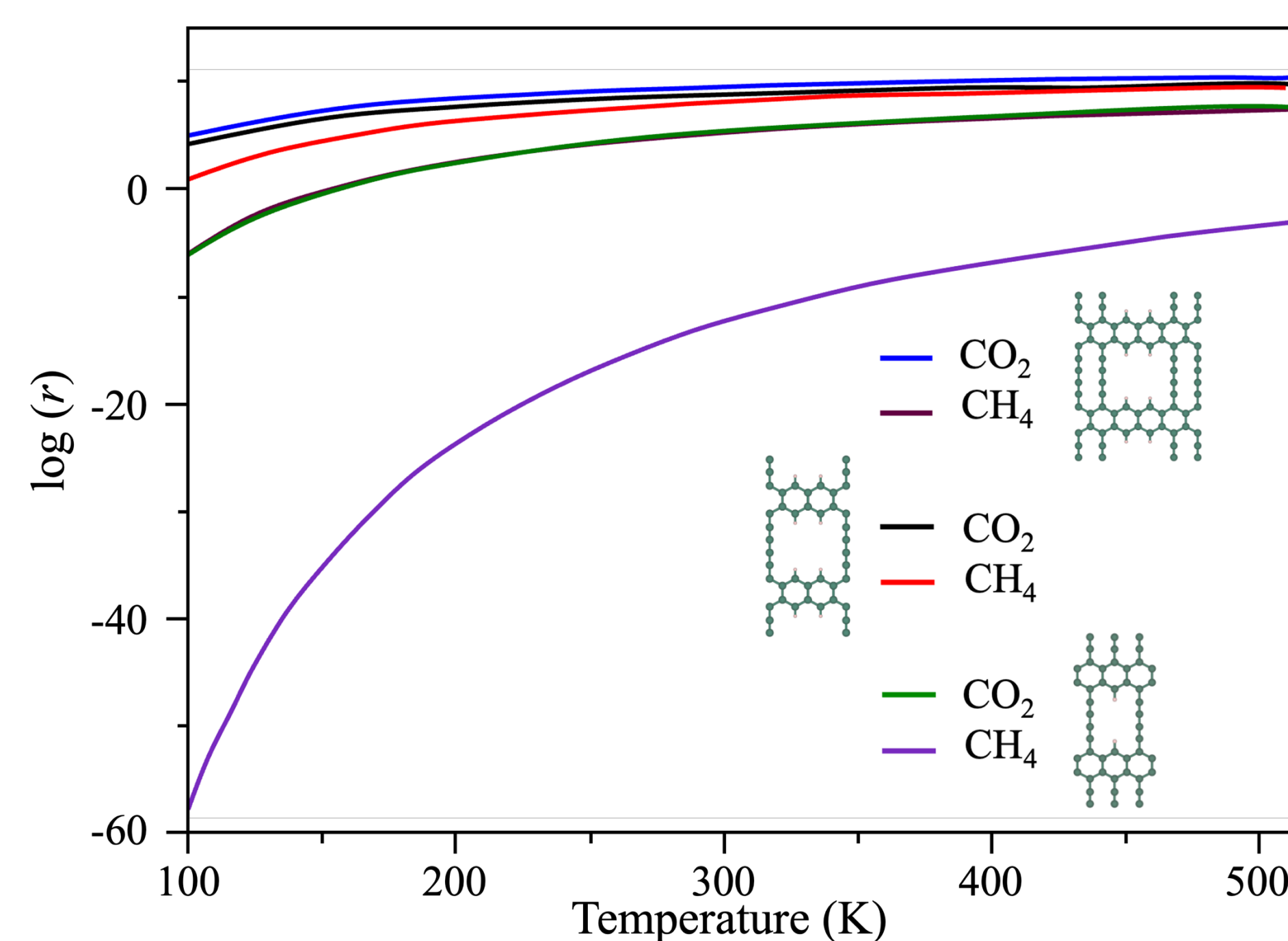


Creating acetylenic **vacancies**

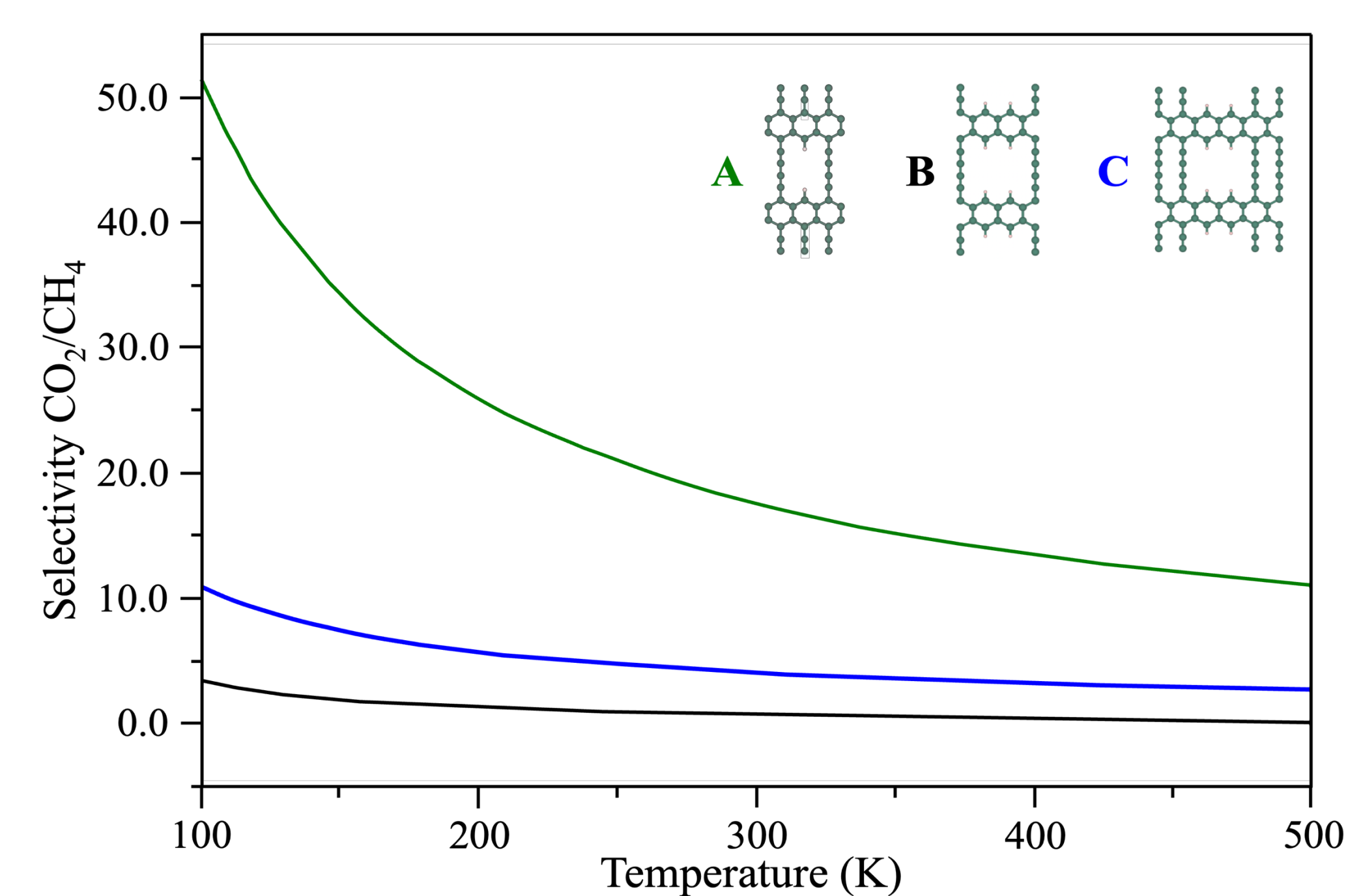
Models



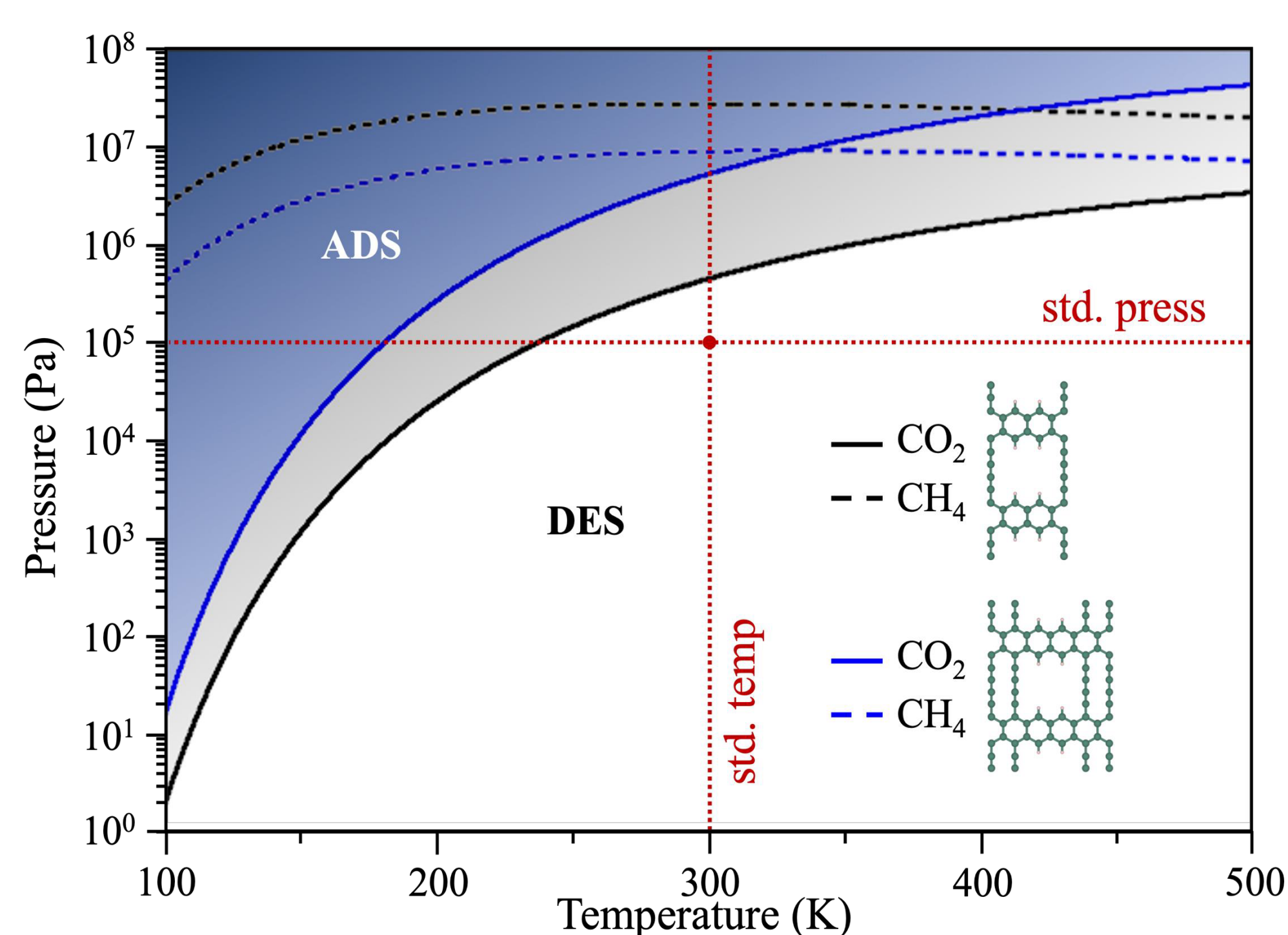
Diffusion Rates



Selectivity



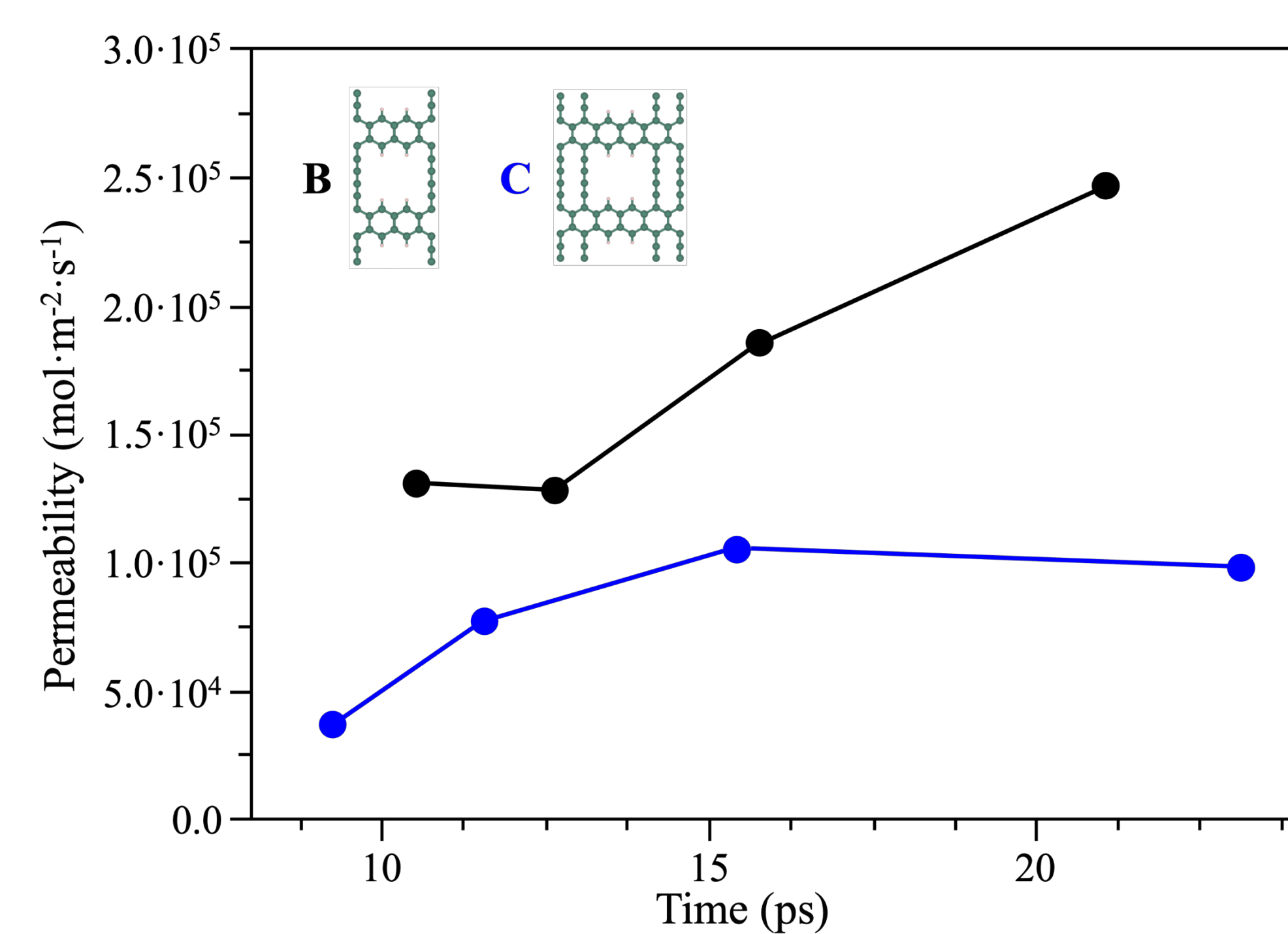
Kinetic Phase Diagrams



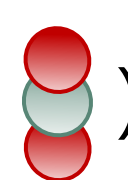
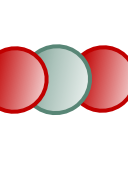
Molecular Dynamics

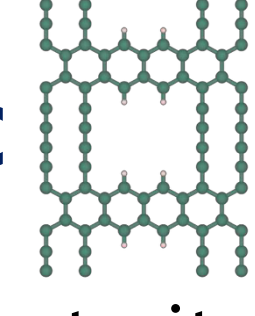
	<i>P</i>	21.05	15.78	12.63	10.52
B	#CO ₂	39	27	24	23
	#CH ₄	1	1	1	1
	<i>P</i>	23.11	15.40	11.55	9.24
C	#CO ₂	20	30	14	12
	#CH ₄	0	0	0	0

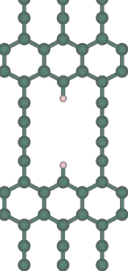
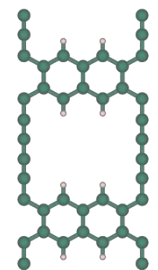
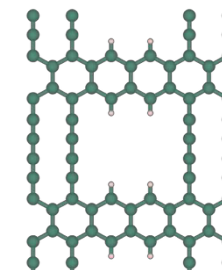
* Pressure in atm

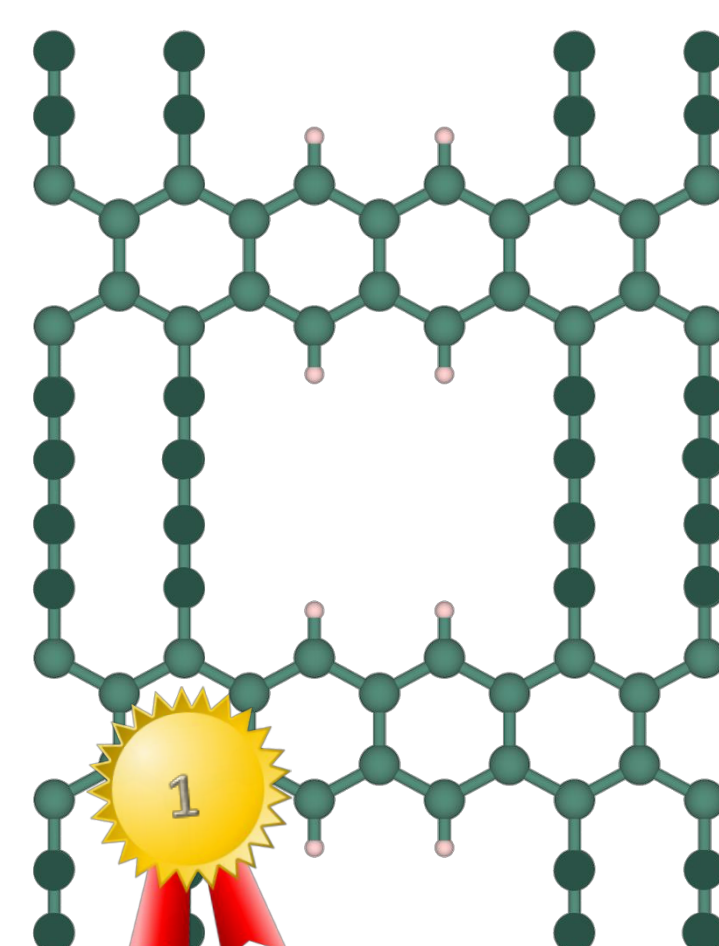


Conclusions

Increasing the pore size resulted in **decreased adsorption and barrier energies** for CO₂ and CH₄. The diffusion barrier energy significantly decreases with larger pores due to lower repulsion in the transition state (TS). **Perpendicular**  diffusion of CO₂ is **energetically more favorable** than **parallel**  diffusion.

Compound C  was found to be **the most suitable for biogas upgrading** due to its higher diffusion rates and selectivity.

Compound A  showed to be **completely impermeable**, CO₂ molecules pass through **Compound B**  and **Compound C** , while CH₄ only passes through **Compound B**.



Acknowledgments



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PID2021-126076NB-I00

TED2021-129506B-C22

PID2022-138180OB-I00

MICIU/AEI/10.13039/501100011033

CEX2021-001202-M



2021-SGR-00079

References

Check out the paper!



...and more about grazynes



F. Viñes, A. Calzada, P. Gamallo, *J. CO₂ Util.* **2023**, *71*, 102459.

1. S. Kamalanihad, F. Viñes, P. Gamallo, *J. Phys. Chem. C* **2019**, *123*, 27140-27149.